

Complexity Aware Recursive Decomposition for Math Reasoning

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Abstract

Large language models often fail on multi-step reasoning due to fixed reasoning strategies that ignore problem specific difficulty. We introduce CARD (Complexity Aware Recursive Decomposition), a framework that predicts problem complexity before generation and adapts decomposition accordingly. Our system comprises MRCE (Multi-dimensional Reasoning Complexity Estimator), a 0.6B Qwen model predicting 30 fine-grained features from question text and a two-stage recursive solver: (1) hierarchical decomposition into K steps based on task profile and (2) per-step thought budget allocation (1, 5-9, or 10 thoughts) via recursive MRCE profiling. Evaluated on three reasoning models (Qwen3-0.6B, DeepSeek-R1-Distill-Qwen-1.5B, Qwen3-1.7B), CARD achieves 81.4 to 89.2% accuracy on GSM8K while reducing token cost by 2.16× to 2.40× compared to fixed decomposition baselines. On MATH-500, CARD reaches 75.1 to 86.8% accuracy using 1.98× to 6.13× fewer tokens. Our results demonstrate that *preemptive* complexity estimation enables both higher accuracy and significant efficiency gains.

CCS Concepts

• Computing methodologies → Natural language processing; Machine learning; Planning and scheduling.

Keywords

Recursive Decomposition, Complexity Estimation, Complexity-Aware Reasoning, Token Efficiency, Mathematical Reasoning

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1 Introduction

Large language models have made significant progress on multi-step reasoning tasks, but performance drops sharply as problems get harder [6, 11, 12]. Prompting methods like Chain-of-Thought (CoT) [12], Tree of Thoughts (ToT) [13] and recursive decomposition [10, 14] work by breaking complex reasoning into smaller steps. The problem is these systems use *fixed decomposition strategies*

regardless of how hard the problem actually is. Simple problems get over-processed, while complex ones don't get enough reasoning depth.

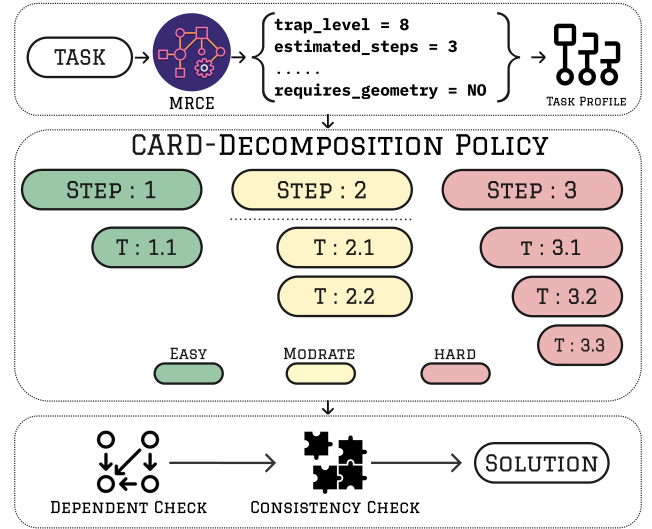


Figure 1: CARD framework architecture and execution flow.

Take two GSM8K problems: "A bakery sold 23 loaves. If each costs \$4, what's the total?" versus "A train travels at varying speeds through 5 segments with different conditions, calculating fuel consumption per segment based on speed and load..." The first needs one multiplication (23×4). The second requires multi-step decomposition with dependency tracking across sub-calculations. Yet methods like Recursive Decomposition of Logical Thoughts (RDoLT) [10] apply the same 3-tier/3-thoughts recursive decomposition to both, burning ~2,000 tokens on the trivial case while the complex one might need deeper decomposition. This mismatch between fixed procedures and variable problem complexity causes cascading errors in long inference chains and wastes tokens.

Current decomposition methods lack *preemptive* complexity awareness. They either use predetermined strategies (ToT uses fixed beam search, RDoLT uses fixed 3-tier depth) or react to failures (ADaPT [8] decomposes only after detecting errors). No prior work estimates multi-dimensional problem complexity arithmetic operations, reasoning depth, semantic difficulty and model-specific failure probability *before* starting decomposition. Without this, there's no way to allocate computational resources proportional to actual problem requirements.

We propose CARD (Figure 1), a framework that predicts fine-grained reasoning complexity before generation and adapts its

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two-stage decomposition accordingly. CARD has two components (1) *MRCE* (Multi-dimensional Reasoning Complexity Estimator), a 0.6B Qwen model fine-tuned to predict 30 features from question text arithmetic structure (operations, steps), reasoning depth (variables, entities, estimated steps), semantic difficulty (trap level, inherent difficulty) and calibrated failure probabilities for 1.5B/7B-class models; and (2) a two-stage adaptive solver that first decides whether to decompose based on overall complexity ($\rho \geq 3.0$) then recursively allocates per-step thought budgets (1, 5-9, or 10 thoughts) based on sub-problem complexity. This allows graduated allocation: 367 tokens for trivial problems versus 1,897 tokens for extremely hard problems, compared to RDoLT’s uniform ~2,000 tokens. Our contributions are:

- 1 A multi-dimensional reasoning complexity estimator comprising a 0.6B parameter model that extracts 30 linguistic and structural features from question text to predict decomposition requirements (subsection 3.1).
- 2 An adaptive recursive solver that conditions decomposition depth and branching strategy on predicted complexity scores, terminating recursion when estimated sub-problem difficulty falls below solvability thresholds (subsection 3.2).
- 3 Empirical evaluation on three state-of-the-art small reasoning models (Qwen3-0.6B, DeepSeek-R1-Distill-Qwen-1.5B, Qwen3-1.7B) demonstrating that CARD improves accuracy by 1.4–2.2 points while reducing token usage by 1.71× to 5.74× on GSM8K and MATH-500 (section 4).

2 Related Work

Decomposition-based prompting has become important for complex reasoning tasks. Khot et al. [7] proposed Decomposed Prompting, breaking tasks into modular sub-problems with specialized prompts. Zhou et al. [14] introduced Least-to-Most prompting that generates steps explicitly. Yao et al. [13] developed Tree of Thoughts (ToT) for structured exploration, though this adds significant computational cost. RDoLT [10] applies recursive decomposition with a fixed 3-tier depth. Similarly, in the training domain, [9] introduced Curriculum Guided Adaptive Recursion (CGAR), which dynamically scales recursion depth based on instance complexity to accelerate convergence. ADaPT [8] uses reactive decomposition triggered when execution fails. Our approach differs: CARD performs *pre-emptive* complexity-aware decomposition before generation starts. Recent work explores predicting task difficulty to guide reasoning. Kang et al. [6] analyzed how decomposition benefits change with task structure. Wu et al. [1] trained ComplexityNet to predict whether a model will succeed, cutting resource use by 90%. Kim et al. [3] used linear probes to estimate knowledge without generation. These methods work post-hoc or reactively. CARD instead integrates fine-grained 30-feature complexity profiling *before* decomposition happens. Wang et al. [4] proposed token-budget-aware reasoning. We combine preemptive estimation with recursive control, beating both fixed decomposition [10] and post-hoc adaptation [8].

3 Method

Given a reasoning task q , CARD extracts a complexity profile $\mathbf{p}_q \in \mathbb{R}^{30}$ then executes two-stage decomposition: (i) step decomposition and (ii) thought generation with complexity-dependent budgets.

3.1 MRCE: Multi-dimensional Reasoning Complexity Estimator

Let q denote the raw question text and $\mathbf{p}_q = [f_1, f_2, \dots, f_{30}]^T$ a feature vector encoding fine-grained complexity attributes. MRCE is a Qwen3-0.6B base model fine-tuned with LoRA on $\mathcal{D} = \{(q_i, \mathbf{p}_i)\}_{i=1}^{100k}$ problem-profile pairs to learn:

$$\mathbf{p}_q = \text{MRCE}_\theta(q; \mathcal{D}) \in \mathbb{R}^{30} \quad (1)$$

where θ are LoRA adapters (rank $r = 128$, $\alpha = 256$) optimized via AdamW with learning rate $\eta = 2 \times 10^{-4}$ over $T = 950$ iterations, adapting the final 16 of 28 transformer layers. Training pairs were generated by prompting GLM-4.6 to extract all 30 features (arithmetic ops, trap_level, estimated_steps, etc.) from question-solution pairs in open-source math benchmarks; MRCE learns to predict these from question text alone.

The profile $\mathbf{p}_q$ breaks down into four groups: Arithmetic structure, Reasoning depth (vars, entities, estimated_steps), Semantic difficulty (trap_level, difficulty) and Calibrated failure probabilities ($f_{1.5B}, f_{7B}$). We define a normalized complexity score:

$$\rho(q; \mathbf{p}_q) = \frac{1}{Z} (w_s \cdot \text{steps} + w_t \cdot \text{trap_level} + w_d \cdot \text{difficulty} + w_f \cdot f_{7B}) \quad (2)$$

where $\mathbf{w} = [w_s, w_t, w_d, w_f] = [0.30, 0.25, 0.25, 0.20]$ and $Z = \sum_i w_i$ ensures $\rho \in [0, 10]$. Fig. 2 shows MRCE’s training convergence and prediction accuracy across feature groups (not all features).

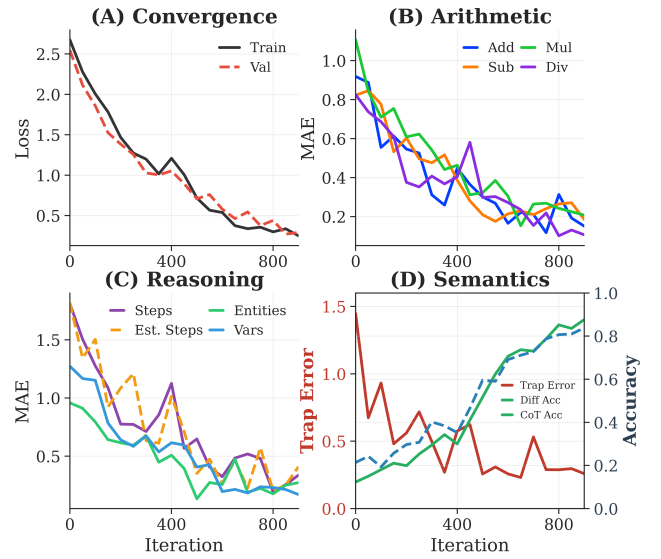


Figure 2: MRCE training convergence and accuracy.

3.2 CARD: Two-Stage Recursive Decomposition

CARD operates in two stages (algorithm details in Fig. ??). **Stage 1: Step-Wise Hierarchical Decomposition.** For tasks with $\rho(q) \geq 3.0$, CARD breaks q into K sequential steps $S_q = \{s_1, \dots, s_K\}$ where K = estimated_steps($\mathbf{p}_q$) is predicted by MRCE.

Stage 2: Complexity-Adaptive Thought Generation. For each step s_i , CARD computes $\rho_i = \rho(s_i; \text{MRCE}_\theta(s_i))$ and assigns thought

CARD Algorithm

```

# Input: q, Params: tau_dec=3.0, tau_low=3.0,
#           tau_high=6.0, alpha=1.5, B_max=10
p_q = MRCE_theta(q); rho_q = Complexity(p_q)
if rho_q < tau_dec: return LLM(q)

K = p_q.estimated_steps
S_q = Decompose_LLM(q, K, p_q) # Step decomp
G = BuildDependencyGraph(S_q); C = [q, G]

for i in range(1, K+1):
    rho_i = Complexity(MRCE_theta(S_q[i]))
    # 3-tier budget: Easy/Moderate/Hard
    B_i = 1 if rho_i < tau_low else (
        ceil(alpha*rho_i) if rho_i < tau_high
        else B_max)
    T_i = {LLM(S_q[i], C, p_si) for j in range(B_i)}
    sol_i = SelectBest(T_i)
    C.append([S_q[i], sol_i])

return Aggregate(C)

```

budget B_i :

$$B_i = \begin{cases} 1 & \text{if } \rho_i < 3.0 \quad (\text{Easy}) \\ \lceil 1.5 \cdot \rho_i \rceil & \text{if } 3.0 \leq \rho_i < 6.0 \quad (\text{Moderate}) \\ 10 & \text{if } \rho_i \geq 6.0 \quad (\text{Hard}) \end{cases} \quad (3)$$

The three-tier discretization emerges from MRCE’s training distribution rather than manual tuning: GLM-4.6 annotations include a continuous "recommended_thoughts" feature, which we threshold via Eq. 3. Ablations (Table 4) show uniform and continuous budgeting underperform this MRCE-guided allocation. For each step s_i , we generate B_i candidate thoughts via context $C_i = q \oplus G \oplus \bigoplus_{j=1}^{i-1} [s_j \oplus \text{sol}(s_j)]$ aggregating the problem, dependency graph and prior solutions.

4 Experiments and Results

We evaluate CARD on GSM8K [2] and MATH-500 [5] using Qwen3-0.6B, DeepSeek-R1-Distill-Qwen-1.5B and Qwen3-1.7B with MRCE (Qwen3-0.6B). **Baselines:** (1) *Zero-shot*: direct prompting without decomposition; (2) *CoT* [12]: "let’s think step-by-step" prompting; (3) *ToT* [13]: beam-3 tree search with breadth-first exploration; (4) *RDoLT* [10]: fixed 3-tier recursive decomposition with uniform 3 thoughts per step. Tables 1 and 2 show CARD’s dual advantage:

Table 1: GSM8K results across three small reasoning models

Method	Qwen3-0.6B	DS-R1-Distill	Qwen3-1.7B
Zero-Shot	79.2 / 1,687	83.65 / 2,043	87.50 / 1,798
CoT	76.8 / 1,847	81.2 / 2,066	85.1 / 816
ToT (beam-3)	78.4 / 2,934	82.9 / 3,187	86.8 / 1,542
RDoLT (3-tier)	79.8 / 2,156	84.1 / 2,289	87.9 / 1,987
CARD	81.4 / 897	85.8 / 1,043	89.2 / 921

Acc (%) / Tokens. DS-R1: DeepSeek-R1-Distill-Qwen-1.5B

better accuracy with fewer tokens. On GSM8K, CARD improves over zero-shot by +1.7 to +2.2 points while cutting token use substantially. For Qwen3-1.7B, CARD achieves 89.2% at 921 tokens

versus zero-shot (87.5%, 1,798 tokens, 1.95× reduction) and RDoLT (87.9%, 1,987 tokens, 2.16× reduction). DeepSeek-R1-Distill reaches 85.8% at 1,043 tokens (1.96× gain over zero-shot). Even Qwen3-0.6B benefits: 81.4% at 897 tokens versus RDoLT’s 79.8% at 2,156 tokens (2.40× efficiency gain). On MATH-500, CARD shows stronger efficiency gains. DeepSeek-R1-Distill+CARD achieves 85.3% at 1,124 tokens versus zero-shot (83.9%, 2,134 tokens, 1.90× reduction) and CoT (81.8%, 6,449 tokens, 5.74× reduction). The 5.74× gain over CoT comes from CARD preventing overly verbose reasoning chains. Qwen3-1.7B+CARD reaches 86.8% at 1,089 tokens, beating zero-shot (1.71× reduction) and RDoLT (1.98× reduction). ToT achieves 84.9% at 2,047 tokens; CARD surpasses this by +1.9 points at nearly half the tokens.

Table 2: MATH-500 results across three reasoning models.

Method	Qwen3-0.6B	DS-R1-Distill	Qwen3-1.7B
Zero-Shot	73.0 / 1,923	83.9 / 2,134	84.57 / 1,867
CoT	71.2 / 1,923	81.8 / 6,449	83.1 / 1,189
ToT (beam-3)	72.8 / 3,156	83.2 / 8,734	84.9 / 2,047
RDoLT (3-tier)	73.6 / 2,487	84.3 / 6,892	85.2 / 2,156
CARD	75.1 / 1,067	85.3 / 1,124	86.8 / 1,089

Table 3 partitions GSM8K problems (Qwen3-1.7B) into complexity quartiles based on MRCE’s predicted $\rho(q)$, showing how CARD achieves both accuracy and efficiency gains through graduated allocation. For the simplest quartile ($\rho \in [0.8, 2.4]$, 25% of data), CARD allocates just 367 tokens per problem, achieving 94.2% accuracy versus RDoLT’s 89.1% at 1,978 tokens a 5.39× efficiency gain with +5.1 point accuracy improvement. RDoLT’s uniform 3-tier decomposition wastes resources on trivial problems that CARD solves with minimal reasoning. For the hardest quartile ($\rho \in [7.2, 9.8]$), CARD uses 1,897 tokens, deploying deep recursive decomposition with thought budgets $B_i \geq 8$, achieving 71.2% accuracy versus RDoLT’s 68.3% at 2,003 tokens. Here, CARD matches RDoLT’s token expenditure while improving accuracy by +2.9 points, correctly identifying when exhaustive search is necessary. The key pattern appears in mid-complexity ranges (Q2-Q3): CARD adjusts decomposition depth dynamically, allocating 712-1,089 tokens to achieve 89.1-92.4% accuracy, while RDoLT’s fixed strategy provides identical 1,982-1,995 token budgets regardless of actual problem requirements, yielding lower accuracy (84.7-86.2%). This graduated, complexity-proportional allocation explains why CARD’s average 921 tokens outperforms methods using 2.16× more resources.

4.1 Ablation Study

Table 4 examines MRCE’s role through three experiments on GSM8K (Qwen3-1.7B). We first compare prompt-based complexity estimation against fine-tuned MRCE. Basic input-output prompting ("Estimate the complexity of this problem") reaches only 77.4% accuracy at 1,287 tokens it systematically misestimates and triggers too much decomposition on simple problems. Adding 3-shot exemplars helps slightly (79.8%) but pushes token cost to 1,456 due to increased prompt overhead per query. Chain-of-Thought self-assessment ("Think step-by-step about this problem’s difficulty")

Table 3: Token allocation by complexity quartile on GSM8K.

Quartile	ρ Range	CARD		RDOLt (3/3)	
		Acc (%)	Tokens	Acc (%)	Tokens
Q1 (Easy)	[0.8, 2.4]	94.2	367	89.1	1,978
Q2 (Moderate)	[2.4, 4.6]	92.4	712	86.2	1,982
Q3 (Hard)	[4.6, 7.2]	89.1	1,089	84.7	1,995
Q4 (Very Hard)	[7.2, 9.8]	71.2	1,897	68.3	2,003

Table 4: MRCE ablation study on GSM8K (Qwen3-1.7B).

Method / Ablation	Acc (%)	Tokens
MRCE (Ours)	89.2	921
<i>vs. Prompt Estimation</i>		
Vanilla I/O	77.4	1,287
3-Shot	79.8	1,456
CoT Self-Assess	76.1	1,623
<i>Feature Group Removal</i>		
w/o Arithmetic	83.7	1,078
w/o Reasoning Depth	79.2	1,598
w/o Semantic	80.4	1,245
w/o Calibration	84.6	1,124
<i>Individual Feature Removal</i>		
w/o estimated_steps	78.4	1,687
w/o trap_level	82.3	1,298
w/o difficulty	85.8	1,089

performs worst: 76.1% with 1,623 tokens. LLMs lack the introspective calibration to self-assess complexity accurately without specialized training. Fine-tuned MRCE, by contrast, achieves 89.2% at 921 tokens, confirming the need for dedicated complexity modeling with task-specific features.

Feature group ablations show which components matter most. Removing Reasoning Depth features (vars, entities, estimated_steps) causes severe over-decomposition 79.2% accuracy and 1,598 tokens. Without these features, CARD cannot predict how many reasoning steps are required and defaults to conservative deep decomposition. Removing Semantic difficulty features (trap_level, difficulty) drops performance to 80.4% at 1,245 tokens; CARD fails to identify conceptually challenging problems that need careful reasoning. Removing Calibrated failure probabilities ($f_{1.5B}$, f_{7B}) yields 84.6% at 1,124 tokens. Model-specific calibration improves allocation, but the effect is smaller. Individual feature ablations confirm estimated_steps is most important (10.8 point loss, 1,687 tokens), followed by trap_level (6.9 points, 1,298 tokens) and difficulty (3.4 points, 1,089 tokens). Multi-dimensional complexity profiling is necessary for effective adaptive decomposition.

5 Conclusion

We presented CARD, a recursive reasoning framework that replaces fixed-depth decomposition with preemptive complexity assessment.

By estimating difficulty across 30 dimensions via our 0.6B MRCE estimator, CARD adaptively allocates token budgets (from 367 to 1,897 tokens). Across Qwen3 and DeepSeek-Distill models on GSM8K and MATH-500, CARD achieved higher accuracy with $1.71\times$ to $5.74\times$ lower token costs than uniform baselines like RDOLt.

Limitations & Future Work: CARD’s features are currently math-specific; generalizing to code or science requires new feature engineering. Future research will explore (1) self-supervised estimation via confidence calibration, (2) RL-based decomposition policies with efficiency rewards, and (3) MoE routing for multi-model deployment.

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